

Stage 2: Results and Analysis

Single-hop Causal Head Map in OLMo-2-7B

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1 Run and validation

Run `stage2_v1` completed successfully as Slurm job 888691 in 1h14m42s with exit code 0. Three persistent replicas used six GPUs in total, with two GPUs per replica. The run used the base `allenai/OLMo-2-1124-7B` model, revision `7df9a82518afdecae4e8c026b27adccc8c1f0032`, and the same data, behavioral baseline, discovery split, and Stage-1 artifacts as the observational scan.

All mandatory gates passed independently on all three replicas. The behavioral runner drift, self-patch drift, self-attribution effect, and corrupted-input embedding endpoint drift were zero on the checked examples. The gate also covered shared-answer-prefix cases and exact versus gradient previews. These checks establish consistency of this implementation on the tested inputs; they do not establish scientific usefulness of any head.

The output is complete: all 178 clean/corrupted discovery pairs have finite screen and extension files; 40 pairs have exact effects for all 1,024 heads; and all 178 pairs have exact effects for the 92-head extension set. The raw arrays reproduce `head_effects.csv`. Model, data, baseline, Stage-1, and code hashes match the saved manifest.

Extension v2. After the Stage-1 test-only RI extension replaced the candidate list, a targeted second run, `stage2_v2_extension`, added exact measurements without recomputing or modifying `stage2_v1`. Slurm job 912879 completed in 32m51s with exit code 0 on the same node, three replicas, six GPUs. It measured 56 heads at all prompt positions on the 138 pairs outside the 40-pair common subset (7,728 forwards) and 69 heads at the final prompt position only on all 178 pairs (12,282 forwards). The head lists were fixed beforehand by a CPU coverage audit of the first run (`results/stage2_v2_coverage/`). All three replicas passed the original gate plus four new checks: final-position self-patch drift 0; patching corrupted activations at all positions before the first differing token changed the metric by 0 and the activations themselves differed by 0 there (the prefix is identical under causal attention); and the six probe heads reproduced the saved first-run `exact_delta` on family 0 with maximum deviation 0.0. The new all-position values on the common subset therefore coincide bit-for-bit with the first run, and the two runs are pooled below without adjustment.

2 Intervention and outcome measure

For each family and fact order, let x_c be the clean prompt and x_r its token-aligned corrupted twin. The corruption swaps the two answer-side names in the facts, so that the clean distractor becomes the corrupted answer. If the two answer strings share a token prefix c , the metric is evaluated at their first divergent token:

$$M(x) = \text{logit}(y_c \mid x, c) - \text{logit}(y_r \mid x, c). \quad (1)$$

The sign is fixed to the clean answer on both inputs. Thus, $M(x_c) > 0$ and $M(x_r) < 0$ indicate that the model tracks the changed fact. Four of the 178 pairs had a non-empty shared answer prefix.

For head h , exact patching replaces its slice of the input to the attention output projection with the activation from the corrupted run at every original prompt position. Shared answer-prefix positions are recomputed. Define

$$\Delta_h = M(x_c; a_h \leftarrow a_h^r) - M(x_c), \quad I_h = -\Delta_h. \quad (2)$$

We report I_h as *causal importance*: a positive value means that the patch moves the metric toward the corrupted answer. It is measured in logits. It is not an accuracy difference, a sequence-likelihood difference, or an additive decomposition of the clean–corrupted gap.

Across the 89 discovery families, averaging the two orders within each family, the mean clean metric was 5.680 (95% family-bootstrap interval [5.338, 6.030]), while the corrupted metric with the clean sign fixed was -5.491 ($[-5.850, -5.141]$). The mean gap was 11.170 ($[10.632, 11.725]$). Every clean pair had positive M and every corrupted pair had negative M . The intervention contrast is therefore behaviorally strong and sign-consistent.

Two patch scopes. Because the twins share the demonstrations and the first test fact, every activation before the first differing token is identical on the two runs; the all-position patch (*Scope P*) is therefore active on the last 43.5 of 281.7 prompt tokens on average. The extension adds *Scope F*: the head is replaced only at the last prompt position, the colon after **Answer**. Scope F is measured for the 69 heads of groups A and B defined below and is reported separately; it is not pooled with Scope P. For every Scope-F forward the run also stored the clean-answer logit, the corrupted-answer logit and the logit of the source (child) name, so that “promote the corrupted answer” and “suppress the clean answer” can be separated.

Intervals in this report are descriptive percentile bootstrap intervals from 20,000 resamples of whole families (seed 20260915), after averaging the two orders. They were computed after observing the Stage-2 results and are not additional pre-registered hypothesis tests.

3 The first-order screen is well calibrated

The first-order estimate contracts the clean-to-corrupted head-output change at all original prompt positions with the gradient of M at the clean run. On the 40-pair exact subset, using the same pairs for both methods, its signed head-ranking Spearman correlation with exact patching was 0.945. The correlation of absolute mean effects was 0.921, and the pair-by-head correlation was 0.903. This passed the pre-registered threshold of 0.7, so the five-step input-embedding IG fallback was correctly not used.

Two further comparisons separate approximation error from family-sampling variation. Comparing the attribution mean over all 178 pairs with the exact mean over only the 40-pair subset gives a lower correlation of 0.698. In contrast, among the 92 extension heads, attribution and exact patching evaluated on the same 178 pairs correlate at 0.990. The lower cross-sample value is therefore not evidence that the screen failed; it mixes approximation error with a change in the evaluated families.

4 Causal map

Table 1 reports the strongest heads in the all-family exact extension. The extension set was selected on the discovery data using RI, attribution, fixed Stage-1 examples, and random controls. The intervals therefore describe effect stability within this run; they are not selection-adjusted confidence statements.

L17H1 is unusually strong: replacing this one head moves the metric by 5.701 logits on average, approximately 51% of the full clean–corrupted gap. Its effect had the expected sign in every family.

Head	RI	Attribution	Exact I_h	Gap fraction
L17H1	.0094	3.362	5.701 [5.294, 6.112]	.510
L18H19	.0182	2.785	3.072 [2.867, 3.276]	.275
L27H6	.0000	2.640	3.071 [2.749, 3.390]	.275
L18H18	.0151	2.226	2.704 [2.503, 2.907]	.242
L21H18	.0096	1.480	1.704 [1.588, 1.822]	.153
L25H17	.0116	1.183	1.267 [1.190, 1.342]	.113
L22H5	.0155	.950	.983 [.904, 1.063]	.088
L16H21	.0240	.695	.925 [.808, 1.049]	.083
L16H1	.0137	.812	.868 [.766, .972]	.078
L21H6	.0067	.662	.702 [.636, .770]	.063

Table 1: Strongest exact effects among the 92 extension heads. Gap fraction is I_h divided by the mean 11.170-logit clean-corrupted gap. It must not be summed across heads because single-head patching effects can interact.

L18H19, L27H6, and L18H18 each account individually for approximately one quarter of the gap, again with the expected sign in every family. These magnitudes establish intervention sensitivity, not four independent shares of a decomposition.

The all-head exact map used a randomly chosen set of 20 whole families, both orders. Its top four heads were the same four shown above, and their effect magnitudes closely reproduced the 89-family extension: L17H1, L27H6, L18H19, and L18H18 had subset effects 5.742, 3.222, 3.080, and 2.768. The large effects are therefore not artifacts of reporting only the all-family extension.

Across all 1,024 heads, the order-0 versus order-1 exact ranking correlation was 0.461. This value is dominated by the many heads whose effects are near zero. Within the selected 92 heads the correlation was 0.734, and among the overall top 25 it was 0.656; 17 heads appeared in both order-specific top-25 lists. All ten heads in Table 1 had positive mean importance in both orders. Thus the main effects are not confined to one layout, while weaker-head ordering is less stable.

5 Historical pooled RI has weak correspondence with causal importance

The Spearman correlation between pooled first-target RI and exact importance over all heads was 0.135 on the random exact subset. A family bootstrap gave a descriptive stability interval of [0.063, 0.168]. The last-target sensitivity score performed similarly ($\rho = 0.132$). These small positive associations do not support RI as a useful causal ranking for this task.

Only one of the exact top-25 heads appeared in the supported RI-first top 25, and only one appeared in the RI-last top 25. By comparison, 17 of the exact top-25 appeared in the attribution top 25. The overlap result agrees with the rank correlations: the gradient screen tracks exact effects, whereas the RI ranking mostly selects different heads.

The historical pooled rule had selected L3H11 and L9H22. On all 89 families, their exact importance was -0.0022 ($[-0.0067, 0.0024]$) and -0.0020 ($[-0.0065, 0.0027]$), with the expected sign in only 47% and 42% of families. Their median importance was -0.0021 , compared with 0.0002 for the 25 random extension controls. These heads therefore have no detectable effect under this single-head, all-prompt-position patching experiment.

This result sharpens the Stage-1 event analysis. L3H11’s RI events were all previous-token attention from the same period/newline token, and L9H22’s events were all self-attention. More than 99% of their summed first-target scores came from demonstrations. Stage 2 now shows that the large

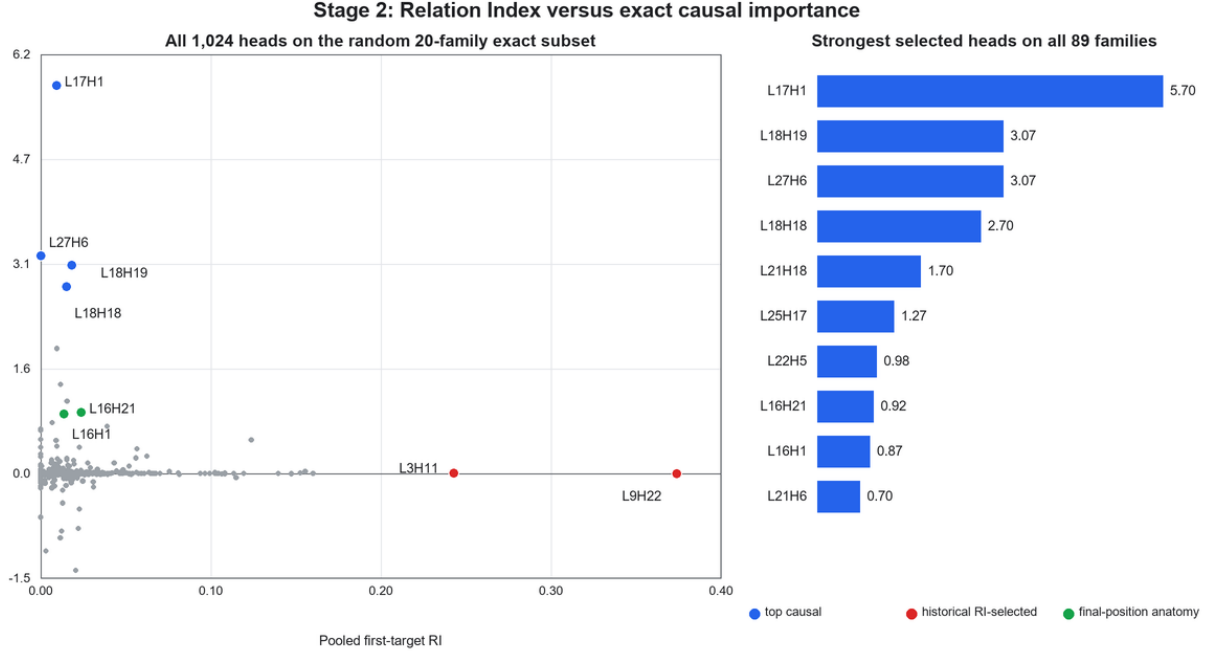


Figure 1: Stage-2 causal map. Left: pooled first-target RI versus exact causal importance for all 1,024 heads on the random 20-family subset. Right: strongest heads among the 92-head exact extension on all 89 discovery families. The historical RI-selected heads lie near zero causal importance, while the largest causal effects have low RI.

observational scores do not translate into movement of the answer metric when their contextual head activations are transplanted.

6 The test-only RI candidates under exact patching

6.1 Coverage and head groups

The Stage-1 extension produced a 59-head discovery union (group A). We also measured the 19 heads with at least one final-position QK event to a test-fact source (group B; 9 overlap with A), and checked the 25 heads with the largest exact effect on the common subset (group C). Of A, 13 heads already had 178-pair exact results in the first run’s 92-head extension (group D); the other 46 were measured in extension v2. All 25 heads of C were already in D, because the first run’s extension included the attribution top 25. In total 148 heads now have Scope-P exact importance on all 89 families.

6.2 Group A: mostly small effects, and the large ones were already known

Eight A heads have $|I_h| > 0.3$ logits (Table 2); all eight were already in D, having entered the first run’s extension through the attribution ranking, not through RI. Among the 46 heads measured for the first time, the largest effects are L8H15 (+0.279, [0.196, 0.361]), L14H23 (−0.212), L16H31 (−0.188), L15H3 (−0.106), L6H24 (+0.101) and L9H16 (+0.095, [0.039, 0.155]). Thirteen of 59 A heads exceed 0.1 logit in absolute value; the remaining 46 are below 1% of the 11.17-logit clean-corrupted gap.

The three heads that the Stage-1 report singled out as interpretive priorities are causally weak. L11H4, the strongest name-gap candidate, has $I_h = 0.004$ logits with an interval that includes zero. L17H5 is slightly negative under both scopes. L9H16 has a small positive all-position effect that

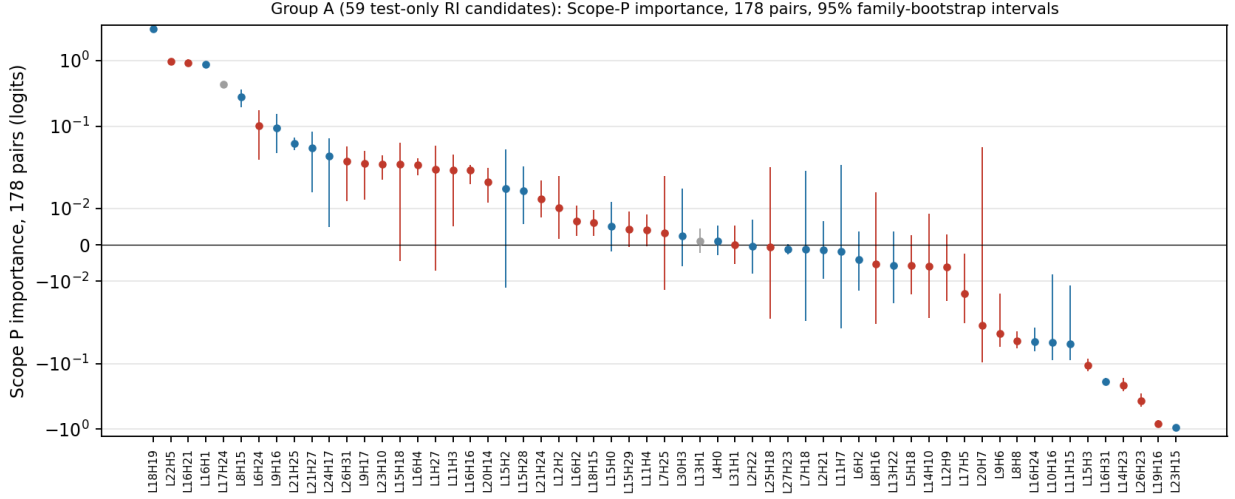


Figure 2: Scope-P importance of the 59 test-only RI candidates on all 178 pairs, with 95% family-bootstrap intervals (symmetric-log axis). Red: positive name gap at the first anchor; blue: non-positive; grey: no supported statistic.

disappears entirely at the final position, so whatever it contributes is written at earlier test positions. Name-gap membership is, if anything, inversely related to effect size inside A: the 36 heads that entered through at least one name-gap ranking have mean $|I_h| = 0.099$ on the common subset, against 0.303 for the 23 heads that entered only through raw-target or word-gap rankings.

6.3 The new RI statistics are negatively rank-correlated with causal importance

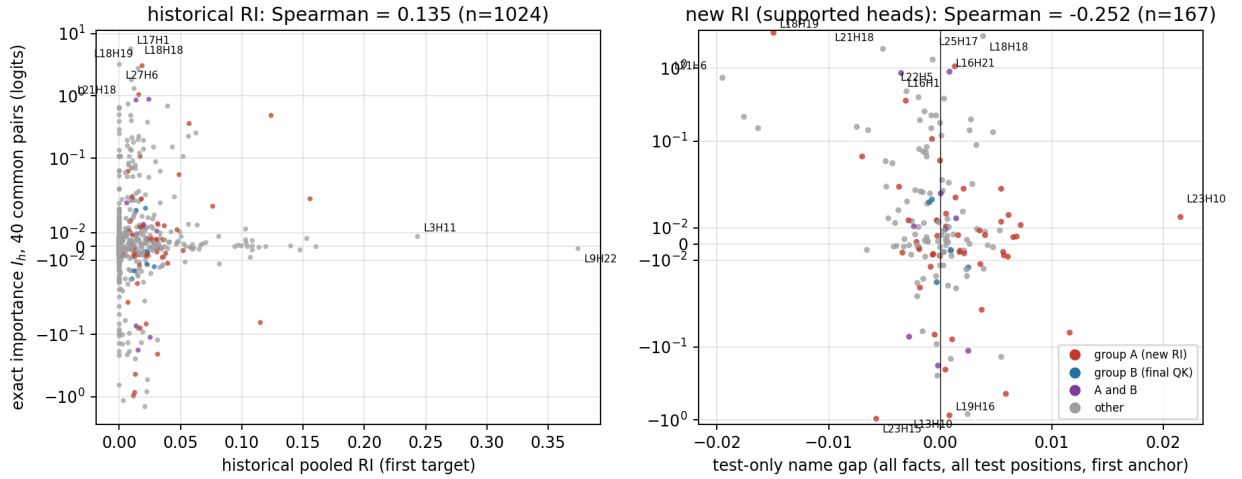


Figure 3: Exact importance on the 40 common pairs (all 1,024 heads, symmetric-log axis) against the historical pooled RI (left) and against the test-only name gap at the first anchor for the 167 heads with at least 20 comparisons in 10 families (right). Colours mark groups A and B.

Table 3 compares RI statistics with exact importance on the common subset, where every head has the same 40 pairs. The historical pooled score has a weak positive correlation (0.135, as reported above). Every test-only name-gap statistic is *negatively* correlated: -0.252 for the main all-facts/all-test first-anchor statistic (family-bootstrap interval $[-0.289, -0.169]$), -0.335 for the query-fact subset, and -0.302 when ranks are taken within layer, so the sign is not a layer artefact. Removing layers 16–18 (-0.278) or the 25 strongest heads (-0.214) does not change it. The raw

Head	Name gap (first)	Scope P I_h	Scope F I_h	F/P	Selection reasons
L18H19	−.0150	3.072 [2.868, 3.275]	3.079	1.00	word gap
L22H5	+.0013	.983 [.901, 1.063]	.984	1.00	target, word gap
L23H15	−.0058	−.947 [−1.062, −.841]	−.953	1.01	target (last)
L16H21	+.0008	.925 [.807, 1.048]	.893	.97	target, name gap (last), word gap
L16H1	−.0035	.868 [.765, .973]	.857	.99	target, word gap
L19H16	+.0008	−.820 [−.911, −.733]	−.829	1.01	name gap, target, word gap
L17H24	—	.433 [.391, .478]	.429	.99	target, word gap
L26H23	+.0059	−.367 [−.458, −.284]	−.369	1.01	name gap, target
L8H15	−.0031	.279 [.196, .361]	−.000	.00	name gap (last), target, word gap
L9H16	−.0007	.095 [.039, .155]	−.001	−.01	name gap (last)
L11H4	+.0065	.004 [−.000, .008]	.000	—	name gap, target
L17H5	+.0061	−.013 [−.024, −.002]	−.059	—	name gap, target, word gap

Table 2: Group-A heads with $|I_h| > 0.3$ (top block, all previously in D) and the three Stage-1 interpretive priority heads plus L8H15 (bottom). Importance in logits on all 178 pairs; negative values mean the corrupted-run activation moves the metric *toward the clean answer*. “Name gap” is the test-only all-facts/all-test first-anchor statistic; — means fewer than 20 events or 10 families.

target statistic is near zero. Top- k overlap tells the same story: the Jaccard overlap between the name-gap top 25 and the exact top 25 is 0.02, identical to the historical score and against 0.52 for the gradient screen.

Statistic (heads with support)	n	Spearman vs. exact I_h (40 pairs)
Historical pooled RI, first target (all heads, zero-filled)	1024	+0.135
Test-only target, all facts / all test / first	172	−0.013
Test-only name gap, all facts / all test / first	167	−0.252
same, ranks within layer	158	−0.302
same, without layers 16–18	144	−0.278
same, positive gaps only	78	−0.102
Test-only name gap, query fact / all test / first	103	−0.335
Test-only name gap, all facts / all test / last	167	−0.091
Test-only name gap, self-attention events only	39	−0.537
Test-only name control mean (other visible names)	167	+0.143

Table 3: Rank correlations between Stage-1 statistics and exact causal importance on the 40 common pairs. Support: at least 20 comparisons in 10 families. All values are exploratory descriptives on discovery families.

Why negative? A copying hypothesis. The name gap subtracts, from the target’s raw-embedding OV score, the mean score of the *other* visible names. A head whose OV circuit copies its current token promotes whatever name is at the current position; at the many events where that name is a control rather than the target, its control score rises and its gap falls. Two observations fit this reading: the control-name mean itself is positively correlated with importance (+0.143; +0.235 with $|I_h|$), and the negative correlation is strongest on self-attention events (−0.537), where the current token is a name. The strongest causal heads (Section 7) act by promoting the retrieved name at the answer position, which is exactly the behaviour a copy-type OV produces. Under this hypothesis the name gap is not merely uninformative: it penalizes the mechanism that carries the effect. The hypothesis is testable from weights alone (the copying score of $W_E W_{OV}^h W_U$ on name tokens, which Stage 2 did not compute) and from the saved events (score of the current-token name at self events); both are Stage-3 items.

7 Where the effects arise: final-position patching

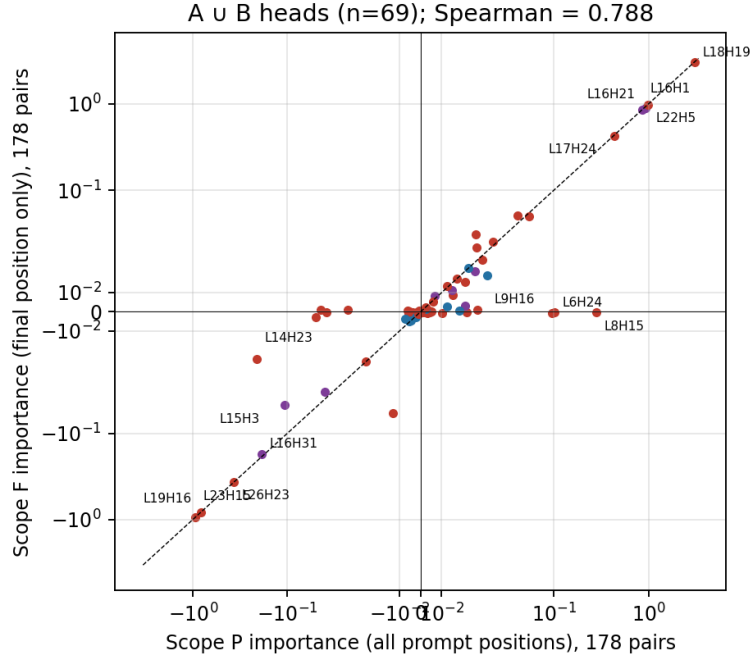


Figure 4: Scope-P (all positions) against Scope-F (final position only) importance for the 69 heads of AUB on all 178 pairs. Dashed line: equality.

For the 69 AUB heads the two scopes correlate at 0.788, but the ratio is bimodal (Figure 5). Every head in layers 16–26 with a measurable effect has I_h^F/I_h^P between 0.91 and 1.09: L18H19 (3.072 vs. 3.079), L22H5, L23H15, L16H21, L16H1, L19H16, L17H24 and L26H23 lose nothing when the patch is restricted to the colon. Their contribution is therefore delivered at the answer position itself. Every head in layers 6–11 with a measurable all-position effect (L6H24, L8H8, L8H15, L9H16, L10H16, L11H15) has a final-position effect indistinguishable from zero; L14H23 and L15H3 are intermediate (0.12 and 0.46). These heads matter only through what they write at earlier test positions, which later components read. Single-position patching does not identify those readers, and the two scopes need not add; but the split is sharp and layer-ordered, which is what a writer-then-reader organization would produce.

The stored logits show *how* the final-position heads act (Figure 6). For L18H19 the patch raises the corrupted-answer logit by 2.88 and lowers the clean-answer logit by only 0.20; for L22H5, L16H21 and L16H1 the pattern is the same (+0.92, +0.90, +0.81 on the corrupted answer against −0.06, +0.01, −0.05 on the clean one). These heads promote the name retrieved on the run that produced the donor activation; they do not suppress the alternative. The source-name logit also rises for L18H19, L16H21 and L16H1 (+0.37, +0.32, +0.33), so the promotion is not perfectly specific to the answer. Three heads have the opposite sign: L23H15 (−0.947), L19H16 (−0.820) and L26H23 (−0.367) *lower* the corrupted-answer logit when their corrupted-run activation is inserted. The consistent reading is that they suppress the name retrieved on their own run, i.e. they behave like the negative name movers of the IOI circuit. Under RI such heads can rank highly (L19H16 and L26H23 entered A through the name gap), because the observational statistic has no sign.

8 Pre-registered audit verdicts

The earlier methodology defined three non-exclusive verdicts. The available evidence supports the following reading.

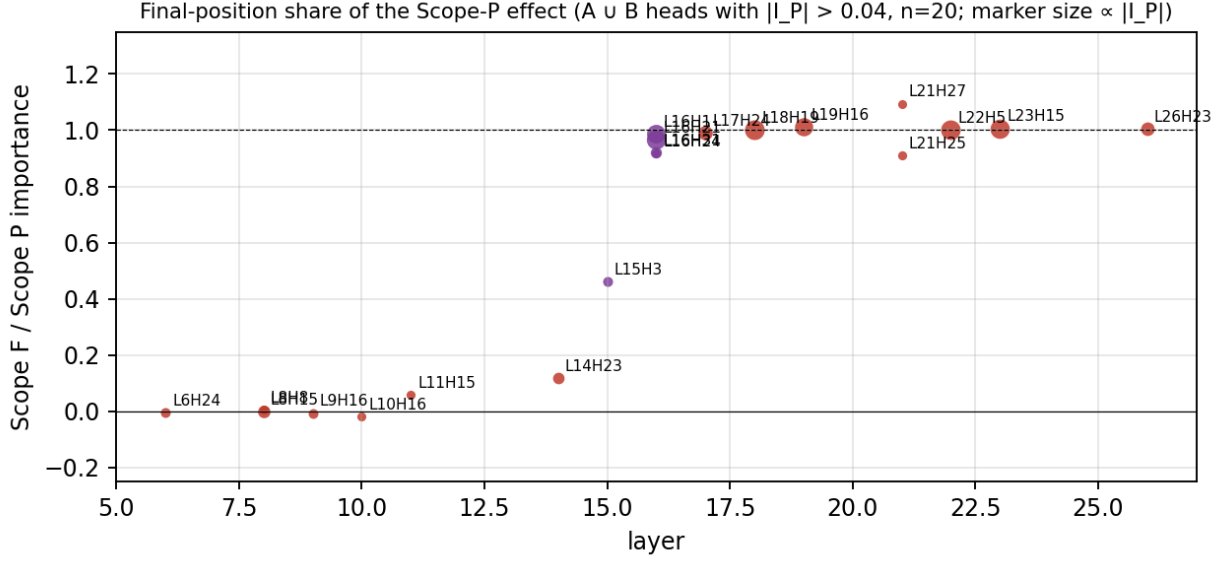


Figure 5: Ratio of Scope-F to Scope-P importance by layer for AUB heads with $|I_h| > 0.04$. Layers 6–11: the effect vanishes when only the final position is patched. Layers 16–26: the final position carries the whole effect.

Incomplete: supported. The lenient 99th percentile of the historical random-token control means was 0.1374. Of the 102 heads in the top decile of positive exact effect, 101 fell below this threshold and 24 had RI exactly zero. Most decisively, L27H6 ranked second in exact importance but had RI zero. Thus a large causal effect can lie below even the lenient RI threshold; this conclusion does not depend on choosing a stricter cutoff after observing Stage 2.

Misleading: descriptively supported. The pre-registered criterion asked whether the median causal effect of RI-passing heads exceeded that of random heads. It did not: the two historical RI-passing heads had median importance -0.0021 , versus 0.0002 for the random controls. The historical RI threshold itself was a descriptive cross-head percentile rather than a calibrated per-head significance test, so this verdict should be reported with that qualification.

Sound: not supported. The methodology described soundness as systematically larger effects for RI-passing heads than for layer-matched non-passing heads, but did not fully operationalize the comparison. Only two heads passed the historical rule, one of which ranked high within its early layer on the 20-family subset but had a tiny effect that vanished on the full 89-family extension. The present data therefore do not support soundness, but we do not turn the underspecified criterion into a new post-hoc significance test.

These verdicts concern the implemented RI adaptation and causal experiment on this dataset. They are not a universal rejection of semantic induction heads or of attention-based diagnostics.

Verdicts under the test-only definition

The same three criteria were re-evaluated with “RI-passing” meaning membership in the 59-head discovery union. Because that union was selected after inspecting Stage-1 results, these are exploratory readings.

Incomplete: supported, more strongly. 88 of the 102 heads in the top decile of exact importance are outside A, including three of the top four (L17H1, L27H6, L18H18). Of the 88, 23 have no test-only QK pass at all and a further 22 fall below the 20-event/10-family support rule, so 45 are misses of the QK stage; 18 are supported with non-positive name gap at both anchors; 25 have a positive gap that did not reach any top 10.

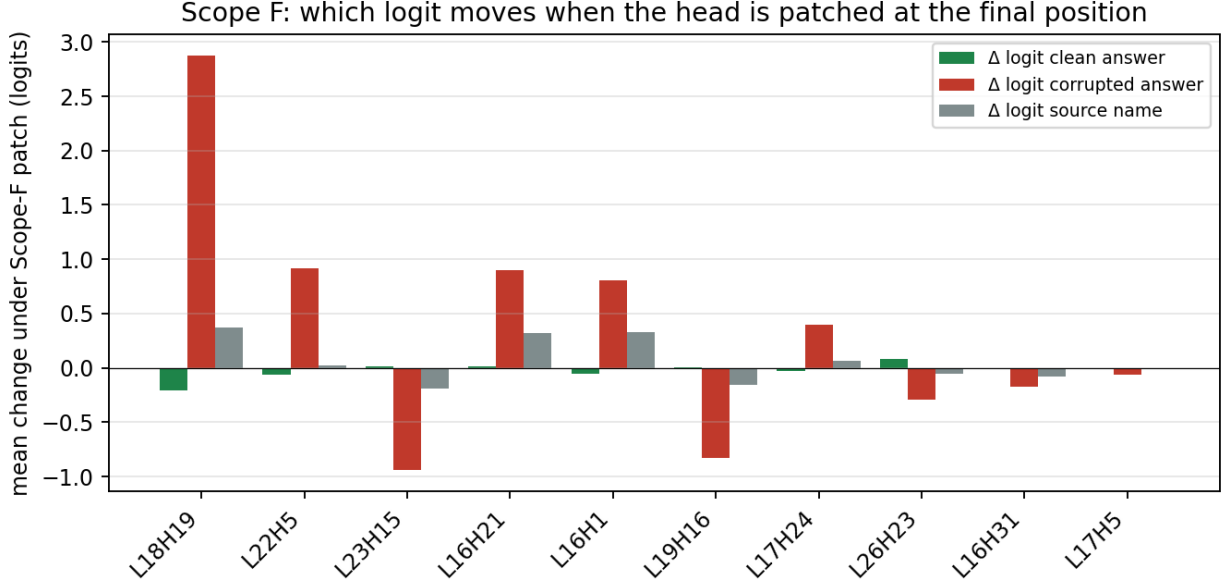


Figure 6: Scope-F patch, ten largest effects: mean change of the clean-answer, corrupted-answer and source-name logits at the first divergent token (89 families, orders averaged).

Misleading: not supported by the letter of the criterion, but the margin is negligible.

The median 178-pair importance of A is 0.0033 logits against 0.0002 for the 25 random controls; the family-bootstrap interval of the difference, [0.0005, 0.0130], excludes zero. The difference is 0.03% of the clean–corrupted gap, and 56% of A heads have positive sign against 60% of the controls. The criterion distinguishes the groups statistically, not scientifically.

Sound: weakly and unevenly.

The pooled layer-matched mean difference between A heads and same-layer non-A heads on the common subset is +0.035 logits ([0.023, 0.047]), but only 12 of 28 layers show a positive difference, and the pooled value is carried by four heads in layers 16, 18 and 22 that the gradient screen had already selected. Layer medians are 0.005 for A against 0.0001 for the rest.

9 How RI misses important heads

The combination of Stage-1 events and Stage-2 effects suggests at least three failure modes.

QK-selection miss. L17H1 had only two scored Stage-1 QK passes across the full scan, yet it was the strongest causal head by a large margin. Its mechanism therefore rarely satisfies the specific condition that the annotated source token be the unique dominant attention target by a factor greater than 2.2. It may distribute attention, retrieve through a different position, or operate on information already moved by upstream components.

Raw-embedding OV miss. L27H6 had 85 scored QK passes but pooled RI zero, despite an exact effect of 3.071 logits. This separates attention eligibility from the embedding-level OV score. A head can be causally important even when $e_j W_V^h W_O^h W_U$ does not promote the annotated target token. Its actual contextual input may carry information absent from the raw token embedding, or its downstream effect may be indirect rather than a direct target-token write.

Sign-blind, copy-penalizing OV. The test-only extension adds a third mode. Its name-gap control was designed to remove heads that merely prefer names in general, but the heads that carry the effect promote names by copying, and the statistic scores that behaviour as a failure whenever the current token is not the target (Table 3). It is also blind to sign: L19H16 and L26H23 rank well

under the gap and turn out to suppress the retrieved answer.

L16H1 and L16H21 provide a fourth informative pattern. Stage 1 found 223 and 230 dominant-attention events, respectively, from the final prompt token to test-fact sources across 76 and 79 families. Stage 2 found exact effects of 0.868 and 0.925 logits. This agreement makes them promising query-position retrieval candidates. However, their low pooled RI shows that the full RI definition still under-ranks them, plausibly because its raw-embedding OV term does not measure their contextual output.

L16H4, which had the smallest raw matched-target calibration p value in Stage 1 and 63 final-position source-attention events, had a positive but much smaller exact effect: 0.0257 logits ($[0.0190, 0.0328]$), approximately 0.2% of the clean-corrupted gap. Dominant source attention and a weak target preference can therefore coexist with only a small causal effect on the answer contrast.

10 Interpretation and next mechanistic questions

Stage 2 establishes a reliable causal screen under a precisely specified intervention. It provides strong evidence that neither the historical pooled RI rule nor the test-only, control-adjusted redefinition is a complete or useful causal selection rule for the model’s single-hop relational behavior. It also identifies a compact set of robust intervention-sensitive heads and, through the final-position scope, a first structural fact about them.

The result does *not* yet identify a circuit. A large effect may reflect semantic relation processing, entity binding, name copying, or another feature that differs between the token-aligned twins. Single-head effects do not establish necessity under redundancy, edges between heads, or additive composition. Scope F narrows *where* the effect enters the residual stream, not what the head computes, and it does not identify which downstream components read the early-layer heads’ output.

What the extension adds is a working hypothesis with a definite shape: a set of layer-6–11 heads whose contribution is written at test-fact positions, and a set of layer-16–26 heads that promote (L18H19, L22H5, L16H21, L16H1, L17H24) or suppress (L23H15, L19H16, L26H23) the retrieved name at the answer position by what looks like copying. The strongest heads of the first run, L17H1, L27H6 and L18H18, still lack a Scope-F measurement and should receive one first.

Stage 3 should therefore begin with L17H1, L18H19, L27H6, L18H18, L22H5, L16H1 and L16H21 as readers, L8H15, L9H16, L6H24 and L14H23 as candidate writers, and L23H15 and L19H16 as negative movers, retaining the random controls. The first questions are:

1. Scope-F for L17H1, L27H6 and L18H18, and a per-position patch profile across the test block for the writer candidates: at which fact tokens does the effect enter?
2. Where does each reader attend on clean and corrupted twins, and does the attention track the query-relevant fact rather than a fixed location?
3. Does the reader’s *actual contextual output* at the colon promote the attended name (copying), and does its weight-space copying score explain the negative name-gap correlation?
4. Path patching from writer to reader: do the layer-6–11 heads supply the readers’ keys or values, or act through a different route?
5. Do pairwise and group interventions reveal redundancy or nonlinear interactions that single-head patches cannot detect, in particular between the positive and negative movers?

Only after these position- and edge-level tests should the components be called a semantic circuit or evaluated for minimal-circuit faithfulness on the held-out validation families.

11 Conclusion

The causal map changes the Stage-1 picture, and the extension sharpens it. The two heads selected by pooled RI have essentially zero exact effect, while several low-RI heads have large, stable, and correctly signed effects. The redesigned test-only RI does not repair this: its three priority heads are causally negligible, its 46 newly measured candidates reach at most 0.28 logits, and its name-gap statistic is negatively rank-correlated with causal importance, most plausibly because the causal heads act by copying names and the control penalizes copying. The first-order screen is strongly validated against exact patching, so none of this is explained by a failed approximation. Within the implemented task and patching scopes, RI in both forms is incomplete as a causal selection rule, and the redefinition is not misleading only in a sense too small to matter. The positive result is structural: the causal heads split by layer into a group whose effect is written at earlier test positions and a group that delivers its entire effect by promoting or suppressing the retrieved name at the answer position. Those two groups, and the edges between them, are what Stage 3 has to test.